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Marchantia). The spores germinate to produce gametophyte.

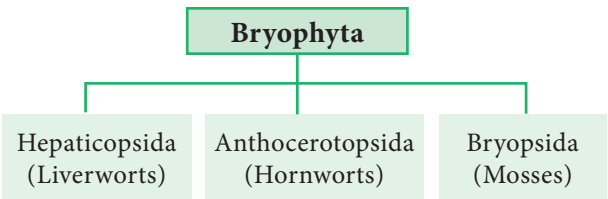
- The zygote, embryo and the sporogonium constitute sporophytic phase. The green long living haploid phase is called gametophytic phase The haploid gametophytic phase alternates with diploid sporophyte and shows heterologous alternation of generation.

2.4.2 Classification of Bryophytes

Proskauer in the year 1957 classified Bryophytes into 3 Classes namely

i. **Hepaticopsida** (*Riccia, Marchantia, Porella* and *Riella*) ii. **Anthocerotopsida** (*Anthoceros* and *Dendroceros*) iii. **Bryopsida** (*Funaria, Polytrichum* and *Sphagnum*).

The outline of the classification is given below



Class: Hepaticopsida

They are lower forms of Bryophytes. They are more simple in structure than mosses and more confined to damp and shady places. They have an undifferentiated thallus. Protonernal stage is absent. Sporophyte is very simple and short lived. In some the foot and seta are absent. Example *Riccia*.

Class: Anthocerotopsida

Gametophyte is undifferentiated thallus. Rhizoids are unicellular and unbranched. Protonemal stage is absent. Sporophyte is differentiated into foot and capsule and seta is absent Example: *Anthoceros*.

Class: Bryopsida

These are higher forms in which the gametophyte is differentiated into 'stem' like and'leaf' like parts and the former showing radial symmetry. Rhizoids are multi-cellular and branched. Protonemal stage is present. Sporophyte is differentiated into foot, seta and capsule. They have a more differentiated structure than liverworts. They often form dense cushions. Example: *Funaria*.

2.4.3 Economic importance

A large amount of dead thallus of *Sphagnum* gets accumulated and compressed, hardened to form peat. In northern Europe peat is used as fuel in commercial scale (Netherlands). Apart from this Nitrates, brown dye and tanning materials are derived from peat. *Sphagnum* and peat are also used in horticulture as packing material because of their water holding capacity. *Marchantia polymorpha* is used to cure pulmonary tuberculosis. *Sphagnum*, *Bryum* and *Polytrichum* are used as food. Bryophytes play a major role in soil formation through succession and help in soil conservation.



Buxbaumia aphylla and *Cryptothallus mirabilis* are saprophytic bryophytes

2.4.4 Marchantia

Class - Hepaticopsida
Order – Marchantiales
Family - Marchantiaceae
Genus - *Marchantia*

Marchantia grows in cool moist shady places. *Marchantia polymorpha* is the common species.